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Anti-inflammatory Effect of Methyl Gallate on Experimental Arthritis: Inhibition of Neutrophil Recruitment, Production of Inflammatory Mediators, and Activation of Macrophages

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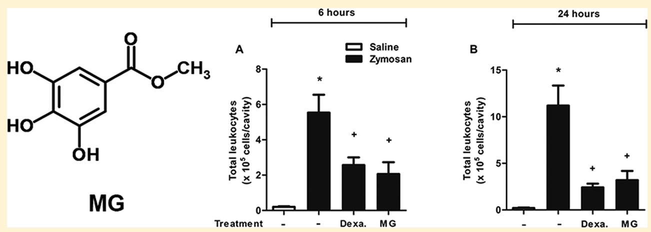
Supporting Information



ABSTRACT: Methyl gallate (MG) is a prevalent phenolic acid in the plant kingdom, and its presence in herbal medicines might be related to its remarkable biological effects, such as its antioxidant, antitumor, and antimicrobial activities. Although some indirect evidence suggests anti-inflammatory activity for MG, there are no studies demonstrating this effect in animal models. Herein, we demonstrated that MG (0.7−70 mg/kg) inhibited zymosan-induced experimental arthritis in a dose-dependent manner. The oral administration of MG (7 mg/kg) attenuates arthritis induced by zymosan, affecting edema formation, leukocyte migration, and the production of inflammatory mediators (IL-1β, IL-6, TNF-α, CXCL-1, LTB_4, and PGE_2). Pretreatment with MG inhibited *in vitro* neutrophil chemotaxis elicited by CXCL-1, as well as the adhesion of these cells to TNF-α-primed endothelial cells. MG also impaired zymosan-stimulated macrophages by inhibiting IL-6 and NO production, COX-2 and iNOS expression, and intracellular calcium mobilization. Thus, MG is likely to present an anti-inflammatory effect by targeting multiple cellular events such as the production of various inflammatory mediators, as well as leukocyte activation and migration.

Arthritis is joint inflammation that can cause edema, pain, and loss of function in the joints. The most common types of arthritis are osteoarthritis, gout, and rheumatoid arthritis (RA).[1,2] The inflammatory process in RA results from the dysregulation of pro-inflammatory cytokines. Tumor necrosis factor (TNF-α), interleukin (IL)-1β, cyclooxygenase-2 (COX-2)-derived pro-inflammatory prostaglandins (PG), and the leukotrienes (LT) produced by lipoxygenases, together with the infiltration of the polymorphonuclear (PMN) and mononuclear leukocytes, are all important elements that highlight the crucial role of the immune system in the pathogenesis of RA.[3,4]

Experimental models of arthritis provide an important approach for evaluating potential anti-inflammatory molecules.[5,6] Zymosan-induced articular inflammation is an arthritis experimental model associated with increases in vascular permeability, neutrophil and mononuclear cell migration, hyperalgesia, and the production of inflammatory mediators, such as TNF-α, IL-1β, IL-6, CXCL-1, PGE_2, and LTB_4.[7−9]

Methyl gallate (MG) is a polyphenol found in various plants and natural products including *Schinus terebinthifolius*, *Rosa rugosa*, and Galla rhois.[10−12] It has been extensively studied because of its antioxidant, antitumor, and antimicrobial activities.[13−15] Regarding antioxidant effects, it has been demonstrated that MG is a free radical scavenger that inhibits lipid peroxidation and shows protective effects against DNA damage resulting from oxidative stress.[16,17] Crispo and coworkers (2010)[18] demonstrated that MG exhibits defensive effects against mitochondrial depolarization, caspase-9 cleavage, and DNA fragmentation in H_2O_2-induced apoptotic conditions. MG has also been shown to delay tumor progression in tumor-bearing animals, displaying an anticancer effect.[19] Indirect evidence suggests an anti-inflammatory activity of MG. Chae and co-workers (2010)[20] showed that MG from Galla rhois inhibited lipopolysaccharide (LPS)-induced IL-6 secretion, nitric oxide (NO) production, and decreasing ERK1/2 phosphorylation in a murine macrophage cell line. However, until now, there have been no descriptions in the literature concerning the *in vivo* anti-inflammatory effect of MG.

Herein, we have demonstrated that oral pretreatment with MG suppresses articular inflammation in the experimental model of zymosan-induced arthritis (ZIA) and elucidated the cellular mechanism of action of this polyphenol.

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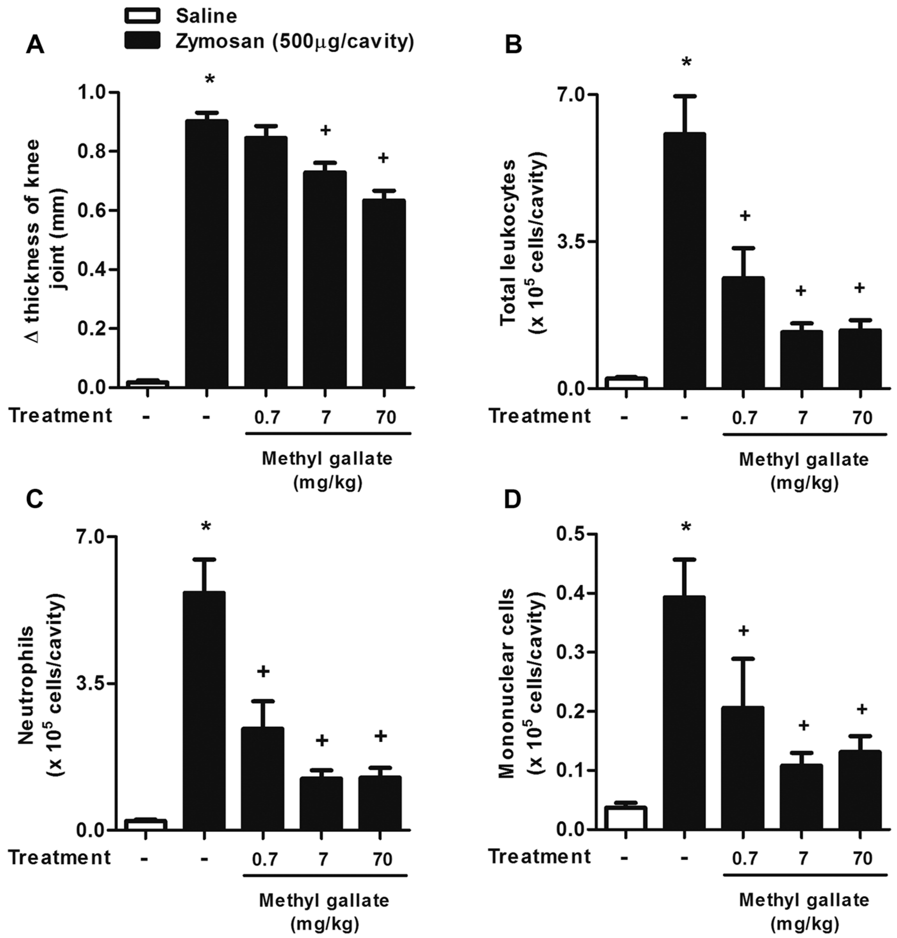

Figure 1. Dose−response analysis for methyl gallate on experimental arthritis induced by zymosan. (A−D) Mice were treated with MG (0.7−70 mg/kg) or vehicle (saline) orally 1 h before i.a. injection of zymosan (500 μg per cavity in 25 μL of sterile saline). The control group was injected with the same volume of sterile saline. Knee joint diameter was evaluated with a digital caliper 6 h after zymosan stimulation (panel A). The numbers of total leukocytes (B), neutrophils (C), and mononuclear cells (D) in the synovial cavity were assessed 6 h after zymosan stimulation. The results are presented as the means ± SEM of seven mice per group per experiment and are representative of two separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the zymosan group (one-way ANOVA followed by Student−Newman−Keuls test)].

RESULTS AND DISCUSSION

Dose−Response Analysis of Methyl Gallate on Cellular Influx and Edema Induced by Zymosan in Mice. The intra-articular injection of zymosan produces a severe and erosive synovitis associated with acute increases in vascular permeability and cell migration, followed by a progressive synovitis.[7,21] In this study, we evaluated the time course of zymosan-induced arthritis in C57BL/6 mice (Figure S1, Supporting Information). In agreement with previous reports,[22,23] the intra-articular (i.a.) injection of zymosan (500 μg/cavity) induced increased knee joint thickness within 6 and 24 h. Here, we also observed that total leukocyte accumulation with a predominance of neutrophils increases within 6 h and remains high until 24 h. Pretreatment with MG (0.7−70 mg/kg, p.o.) was able to inhibit zymosan-induced edema formation at 6 h, whereas no effect was observed for pretreatment with MG at 0.7 mg/kg (Figure 1A). As shown in Figure 1B−D, the leukocyte recruitment was significantly inhibited by the three doses of MG tested, and the same inhibitory effect was observed with doses of 7 and 70 mg/kg. Considering these results, further analyses were performed within 6 and 24 h after stimulation using a dose of 7 mg/kg. Reinforcing the anti-inflammatory effect, the treatment with MG was also able to inhibit two additional models of inflammation, pleurisy and paw edema (Figure S2, Supporting Information). Figure S2A shows that pretreatment with MG (from 1 to 50 mg/kg, p.o.) was able to significantly inhibit paw edema formation (4 h) at doses of 10 and 50 mg/kg. Pretreatment with MG also reduced protein extravasation and neutrophil influx in a pleurisy model at the same doses observed in the paw edema model (10 and 50 mg/kg) (Figure S2B,C). It is likely that MG anti-inflammatory activity involves inhibition of the events that lead to edema formation, plasma exudation, and leukocyte recruitment.

Methyl Gallate Attenuates Articular Inflammatory Response Induced by Zymosan. We investigated the effect of MG at both of the time points discussed above, and we compared its action with that of dexamethasone, an anti-inflammatory reference drug. Oral pretreatment with 7 mg/kg of MG significantly inhibited edema formation at 6 h (24%) and 24 h (49%) after zymosan injection in the knee joint (Figure 2A,B). The reference drug dexamethasone (10 mg/kg, i.p.) also inhibited edema formation at both times (42% and 48%, respectively; Figure 2A,B). The joint swelling in response to zymosan is the result of cascading events that include complement system activation, mast cell degranulation, the generation of 5-lipoxygenase and cyclooxygenase metabolites, and NO production.[8,24−27] It was previously demonstrated that MG inhibited the NO production by LPS-stimulated macro-

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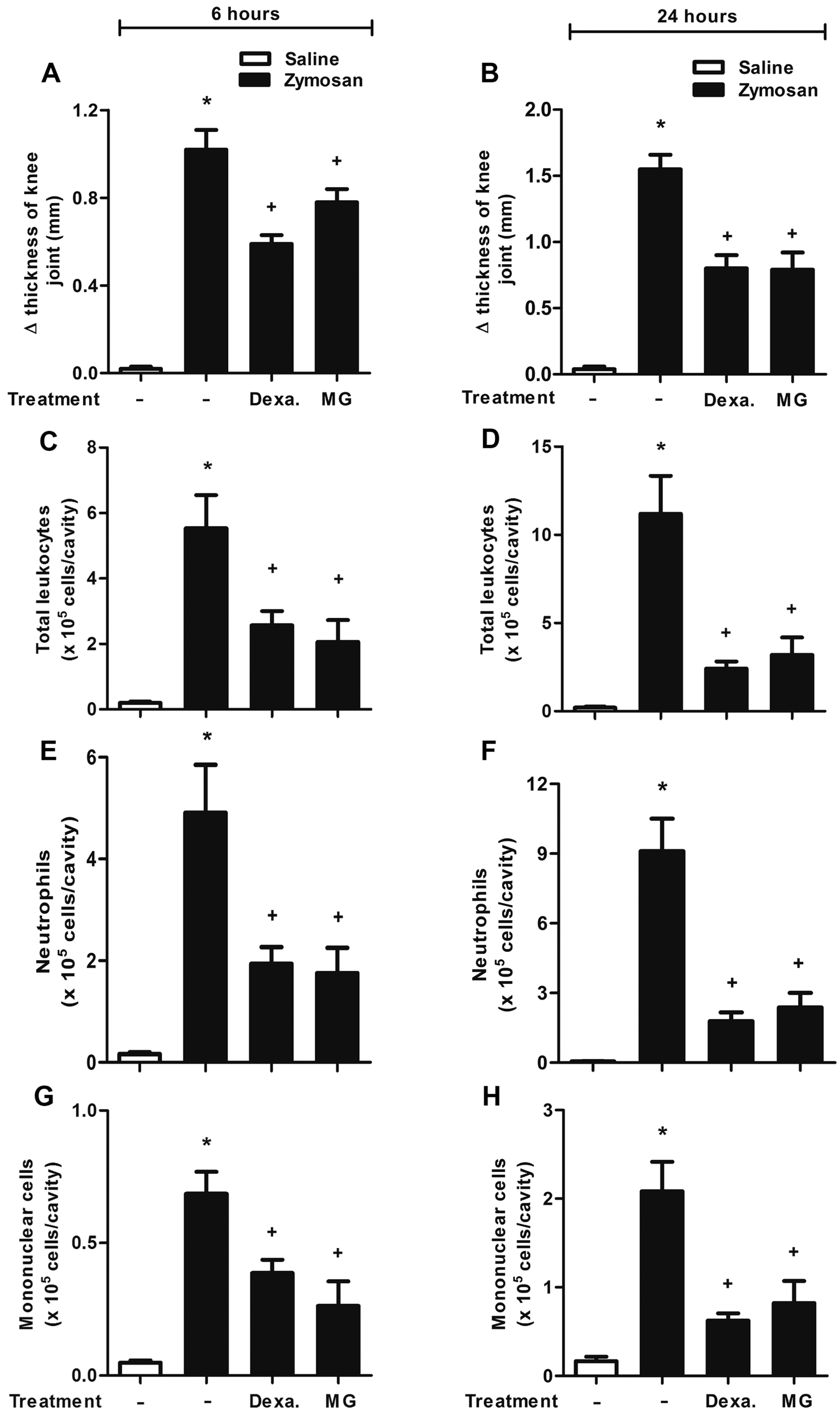

Figure 2. Methyl gallate inhibits edema and leukocyte influx into the articular cavity induced by zymosan. (A–H) Mice were treated with saline, dexamethasone (Dexa, i.p., 10 mg/kg), or MG (p.o., 7 mg/kg) 1 h before i.a. injection of zymosan (500 μg per cavity in 25 μL of sterile saline). The control group was injected with the same volume of sterile saline. Knee joint diameter was evaluated with a digital caliper 6 and 24 h after zymosan stimulation (A and B). The numbers of total leukocytes (C and D), neutrophils (E and F), and mononuclear cells (G and H) are plotted as number of cells $\times 10^5$. Knee synovial cells were recovered 6 and 24 h after zymosan stimulation. The results are presented as the means ± SEM of seven mice per group per experiment and are representative of three separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the zymosan group (one-way ANOVA followed by Student–Newman–Keuls)].

phages.[20] Moreover, MG reduced histamine release by mast cells stimulated with C48/80.[28] In light of these findings, the antiedematogenic effect of MG might be exerted through the inhibition of NO and histamine release during joint swelling.

Pretreatment with MG (7 mg/kg) reduced total leukocyte accumulation in the joint cavity 6 and 24 h after stimulation (Figure 2C,D), particularly by the reduction of neutrophil migration at both times studied (64% and 74%, respectively)

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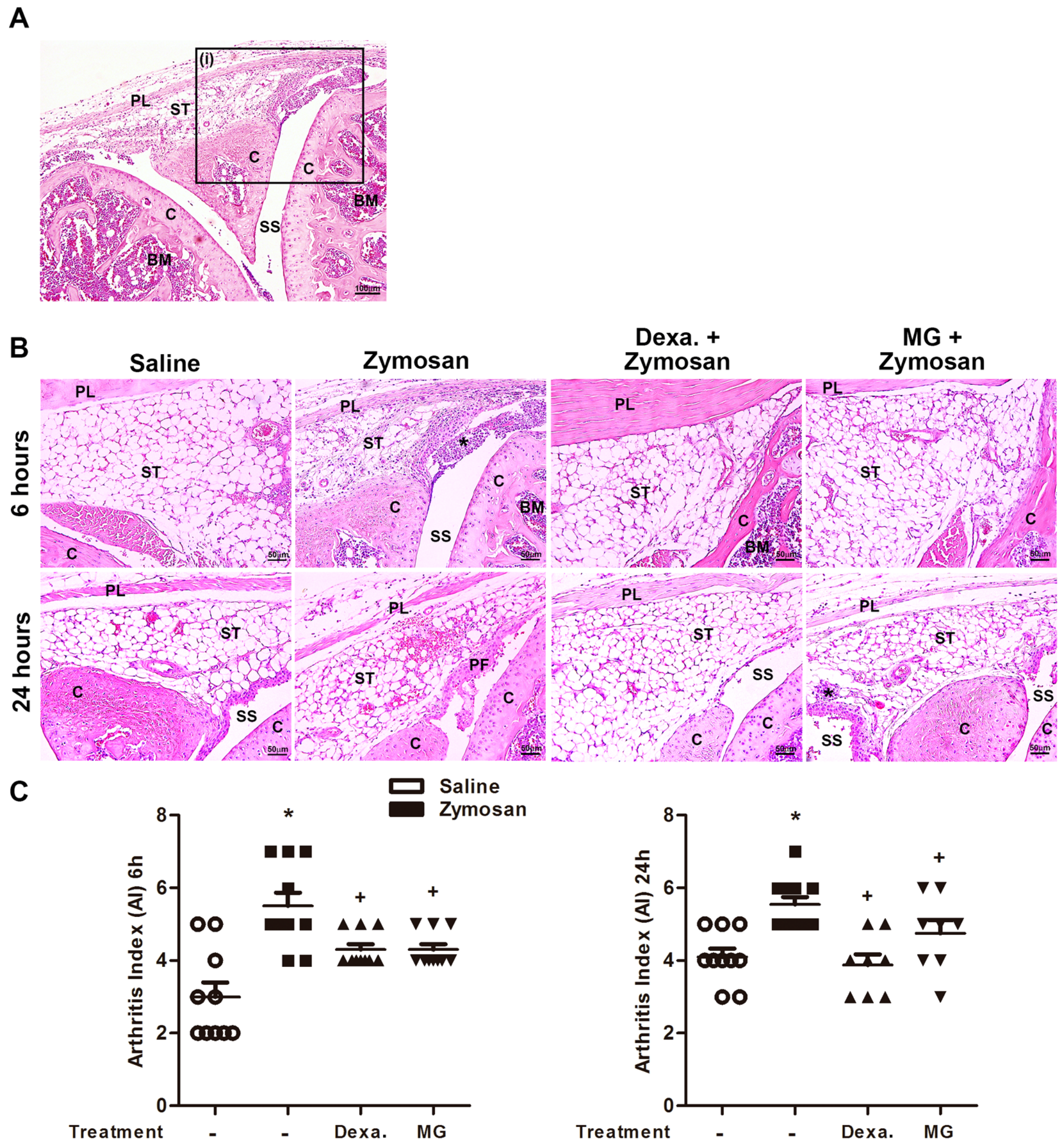

Figure 3. H&E-stained representative histological images from different knee joints injected with saline or zymosan collected 6 or 24 h after stimulation. (A) Image representative of at least six different animals stimulated with zymosan, (i) detached area containing the structures that were evaluated in panel B at higher magnification (10×; bars = 100 μm). (B) Representative images demonstrating histopathology after 6 and 24 h of zymosan stimulus (20×; bars = 50 μm). Saline controls show normal articular cartilage surface and synovial membrane, with absence of inflammatory cells. Non-zymosan-treated group shows the intense infiltration of inflammatory cells (*) and bleeding scattered diffusely in the synovial tissue. A significant reduction in the infiltration of inflammatory cells and local bleeding can be observed in mice pretreated with dexamethasone or MG at both assessed times. (C) Arthritis index (AI) analysis after ZIA onset. The results are presented as the means ± SEM from six mice per group and are representative of two separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the zymosan group (one-way ANOVA followed by Student−Newman−Keuls test)]. C: cartilage; SS: synovial space; PL: patellar ligament; ST: synovial tissue; BM: bone marrow; PF: pannus formation.

(Figure 2E,F). In addition, MG decreased the number of mononuclear cells within 6 and 24 h after stimulus (Figure 2G,H). The positive control, dexamethasone, similarly inhibited zymosan-induced migration of total leukocytes, neutrophils, and mononuclear cells. To confirm these findings, we evaluated the effect of MG on histological joint sections of knee joints (Figure 3). Figure 3A shows representative images of at least six different animals. The highlighted region (i) is shown at higher magnification (20×) in Figure 3B, and it includes part of the synovial tissue, patellar ligament, cartilage, and synovial space. Histopathological analysis of the femorotibial joints of zymosan-stimulated (6 and 24 h) mice revealed intense infiltration of polymorphonuclear cells into the synovium and the periarticular tissues (Figure 3B). The zymosan-stimulated mice showed an increase in the overall arthritis score in comparison to control mice (Figure 3C). Histologic examination of serial joint sections showed that synovial hyperplasia, inflammation, and bleeding were significantly lower in mice treated with MG and dexamethasone, as could be seen in the representative figures for each group (Figure 3B) and their arthritis index scores (Figure 3C).

Neutrophils are the most abundant leukocytes in inflamed joints, and some reports have demonstrated the importance of these cells in the initiation and perpetuation of human RA, as well as in murine models.[29,30] Potential neutrophil effector mechanisms include the release of granules containing

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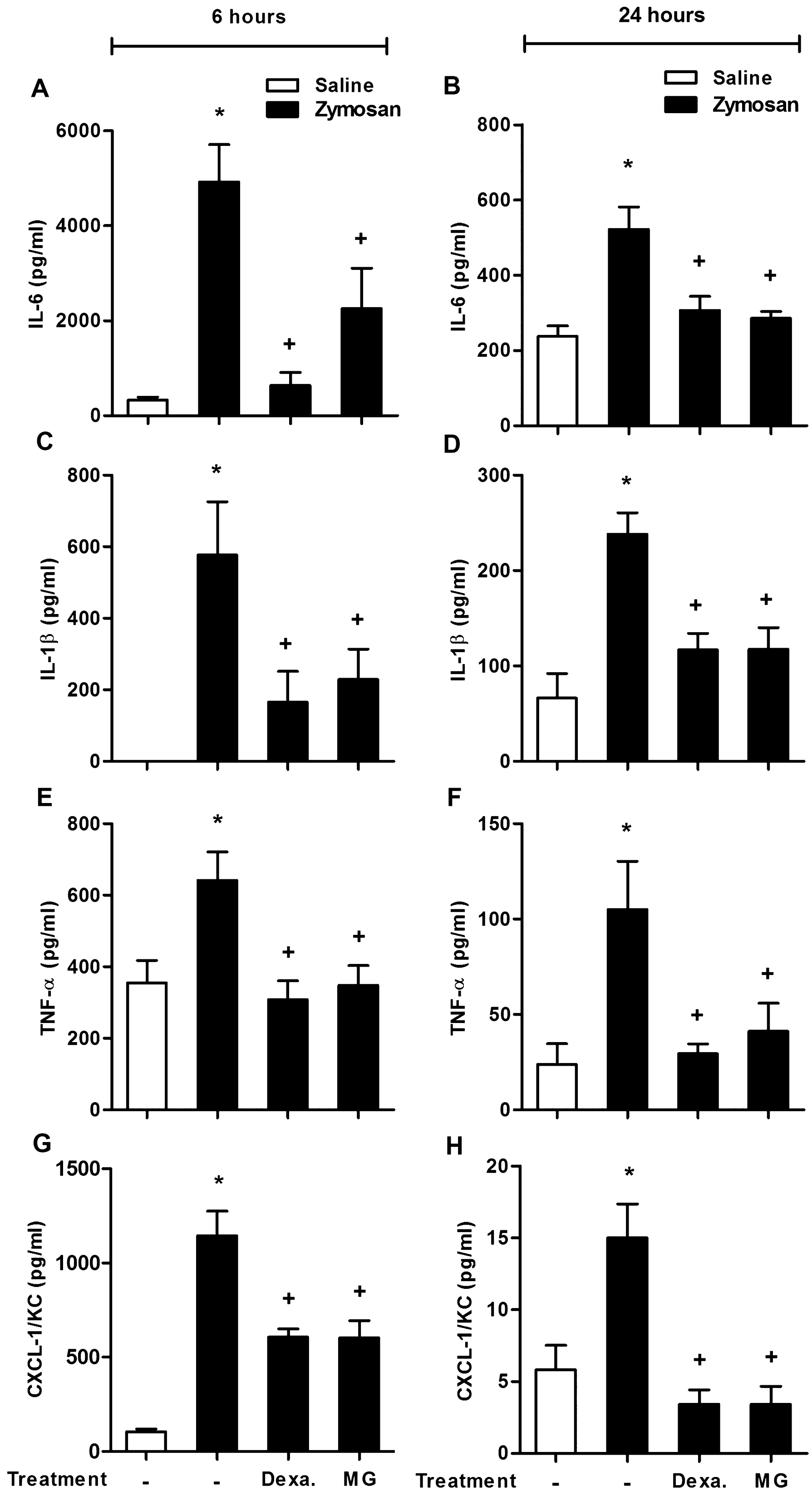

Figure 4. Methyl gallate inhibits inflammatory mediators induced by zymosan. (A–H) Mice were treated with saline, dexamethasone (Dexa, i.p., 10 mg/kg), or MG (p.o., 7 mg/kg) 1 h before i.a. injection of zymosan (500 μg per cavity in 25 μL of sterile saline). The control group was injected with the same volume of sterile saline. Protein levels of IL-6 (A and B), IL-1β (C and D), TNF-α (E and F), and CXCL-1/KC (G and H) were determined in cell-free supernatant of knee-joint synovial washes by ELISA. Analysis was performed 6 and 24 h after zymosan injection. The results are presented as the means ± SEM of six mice per group per experiment and are representative of two separate experiments [$*p \leq 0.05$ compared to the saline group; $^{+}p \leq 0.05$ compared to the zymosan group (one-way ANOVA followed by Student–Newman–Keuls test)].

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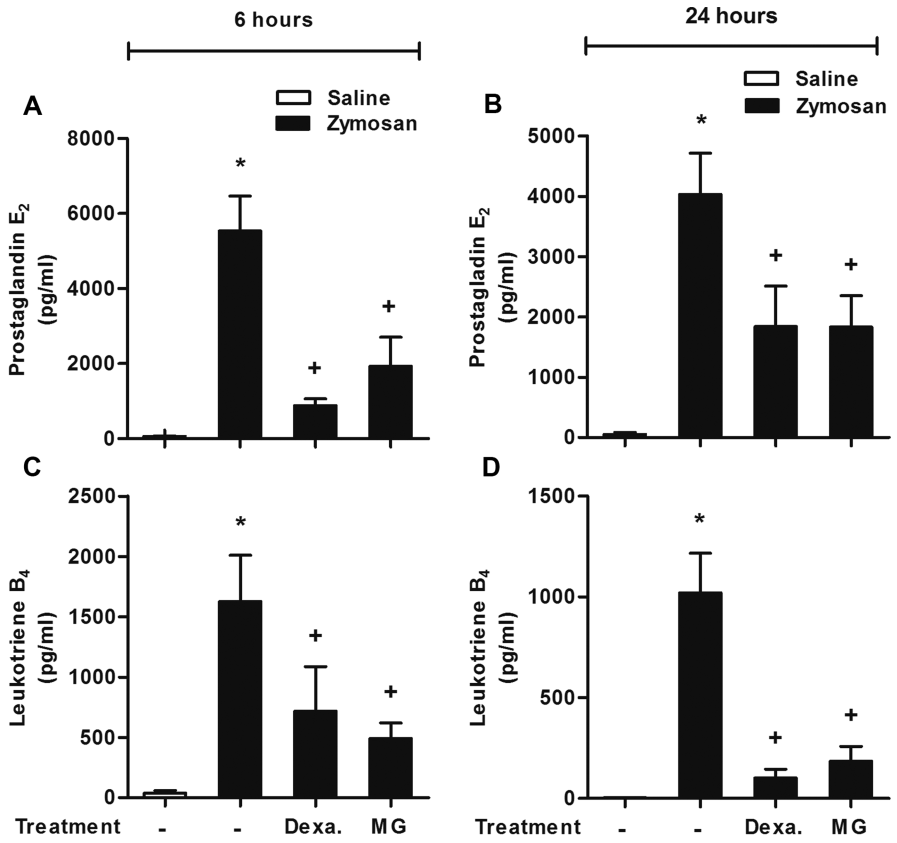

Figure 5. Methyl gallate inhibits lipid mediators induced by zymosan. (A−D) Mice were treated with saline, dexamethasone (Dexa, i.p., 10 mg/kg), or MG (p.o., 7 mg/kg) 1 h before i.a. injection of zymosan (500 μg per cavity in 25 μL of sterile saline). The control group was injected with the same volume of sterile saline. Levels of prostaglandin E_2 (A and B) and leukotriene B_4 (C and D) were determined in cell-free supernatant of knee-joint synovial washes by EIA. Analysis was performed 6 and 24 h after zymosan injection. The results are presented as the means ± SEM of six mice per group per experiment and are representative of two separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the zymosan group (one-way ANOVA followed by Student−Newman−Keuls test)].

degradative enzymes such as myeloperoxidase, elastase, matrix metalloproteinases (MMPs), and collagenase that can cause further damage to the tissue and amplify the neutrophil response.[31,32] Activated neutrophils release pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6 that potentially affect the activities of both neutrophils and other cell types, such as resident mononuclear cells and chondrocytes.[33,34] Therefore, neutrophils play an essential role in joint inflammation, and the modulation of neutrophil functions is considered a potential target for pharmacological intervention in arthritis.[35−37] Our findings suggest that treatment with MG is a plausible approach to control neutrophil recruitment during inflammation, preventing neutrophil-mediated tissue injury.

Methyl Gallate Reduces the Production of Inflammatory Mediators in Experimental Arthritis. Inflammatory signaling pathways in response to zymosan lead to the expression of many pro-inflammatory cytokines, chemokines, and lipid mediators that are detected at early time points in murine models of inflammation.[38,39] In this study, we observed an increase in the production of cytokines, chemokines, and lipid mediators in the mouse joint cavity at 6 and 24 h after zymosan stimulus (Figures 4 and 5). Pretreatment with MG (7 mg/kg) significantly inhibited the zymosan-induced production of IL-6, TNF-α, IL-1β, and CXCL-1/KC, as well as the lipid mediators PGE_2 and LTB_4, at 6 and 24 h, as observed with the reference drug dexamethasone (10 mg/kg, i.p.) (Figures 4 and 5).

Therapeutic strategies that block or antagonize pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6 have been used for the treatment of RA and found to be beneficial in preventing the progression of the disease in patients.[40−42] Experimentally, these cytokines play a critical role in inflammatory hypernociception, drive the production of a neutrophil chemoattractant factor, and increase cell adhesion molecule expression in arthritis experimental models.[43−46] The inhibition of TNF-α and IL-1β receptors (TNFR1 and IL-1R1, respectively) reduced neutrophil migration in the model of antigen-induced arthritis (AIA).[47] Moreover, Sachs and co-workers (2011)[47] showed that the blockade of neutrophil influx with fucoidan decreases TNF-α and IL-1β production. Thus, in AIA, TNF-α and IL-1β are necessary for adequate neutrophil influx, but neutrophils are also essential for the full production of these cytokines. Here, we show that pretreatment with MG was able to reduce the levels of these cytokines at 6 and 24 h after zymosan injection (Figure 4C−F). However, it is necessary to clarify whether MG inhibits the early production of these mediators or whether the reduction in neutrophil accumulation was the reason for the TNF-α and IL-1β decrease observed in our results.

Multiple chemoattractants can directly induce neutrophil migration. Chou and co-workers (2010)[48] demonstrated in the K/BxN murine model of arthritis that LTB_4 was critical for initial neutrophil infiltration. A significant release of LTB_4 1 h after zymosan injection was demonstrated in zymosan-induced

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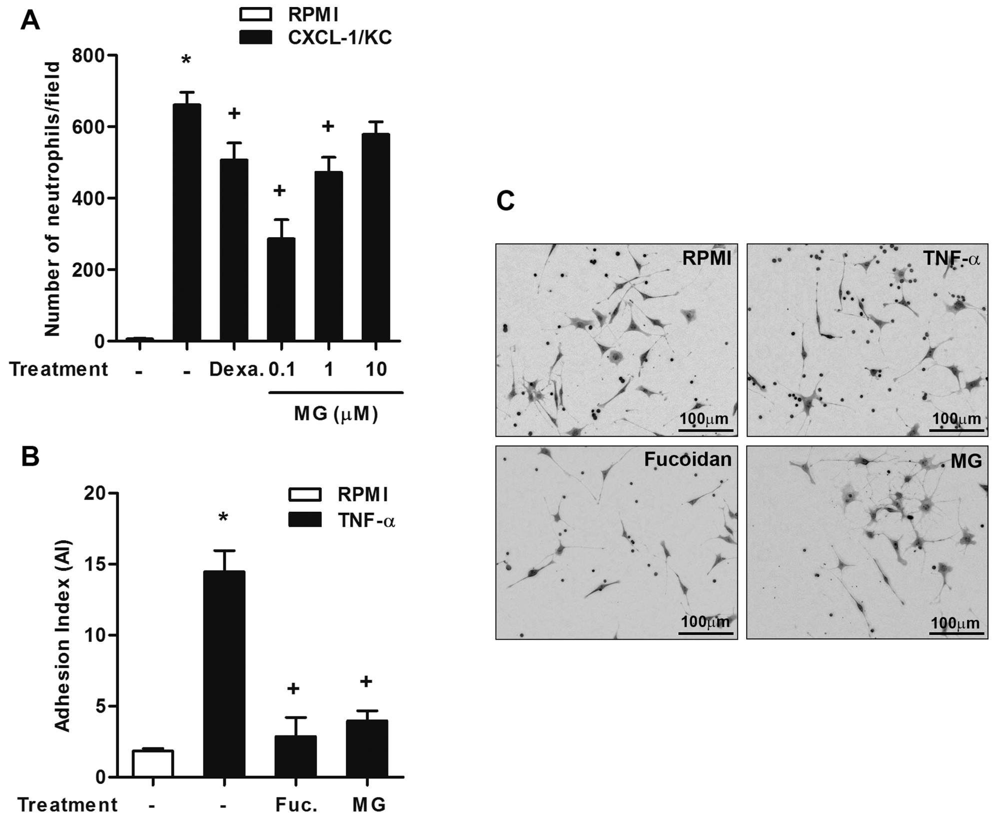

Figure 6. Methyl gallate inhibits murine neutrophil chemotaxis and adhesion. (A) Neutrophils recovered from mouse bone marrow were preincubated for 1 h with medium (RPMI), dexamethasone (Dexa, 50 nM), or MG (0.1, 1, or 10 μM) and allowed to migrate toward CXCL-1/KC (250 nM) for 1 h in a chemotaxis chamber. The migrated cells were counted using light microscopy. The results are expressed as numbers of neutrophils per field per well. (B) Neutrophils were preincubated for 1 h with medium, fucoidan (25 μg/mL), or MG (0.1 μM) and allowed to adhere to previously stimulated tEnd.1 (50 neutrophils/tEnd.1) for 1 h. Fucoidan was used as the positive control. Adhesion was quantified by an association index. (C) Photomicrographs of adherent neutrophils on stimulated tEnd.1 (original magnification 60×; bars = 100 μm). The results are presented as the means ± SEM of quadruplicate wells per group per experiment and are representative of two separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the stimulated group (one-way ANOVA followed by Student−Newman−Keuls test)].

arthritis.[49] Because the neutrophils' influx into the joints in ZIA starts at least after 2 h, it is clear that the LTB_4 production precedes the influx of these cells. However, the treatment with fucoidan or antineutrophil antibody inhibited the zymosan-induced LTB_4 production in the joint 7 h after zymosan stimulation.[9] These data show that although resident cells, such as synoviocytes, might account for the immediate LTB_4 release, neutrophils are responsible for further production of LTB_4.

These LTB_4-recruited neutrophils produce cytokines such as IL-1β, which then act on resident cells, inducing the expression of chemokine ligands for CXCR-1 and CXCR-2 (including CXCL-1/KC).[8,50] These data show the existence of a positive-feedback loop involving lipid−cytokine−chemokine cascades that drive neutrophil recruitment and the development of arthritis.[51] In this study, we demonstrated that MG pretreatment reduces the increases in LTB_4, CXCL-1/KC, and IL-1β in knee joint fluids after zymosan stimulus (Figures 4C−H and 5B,C), as well as the zymosan-induced massive neutrophil migration. It is very likely that MG modulates lipid−cytokine−chemokine cascades and impairs neutrophil influx into the inflammatory site.

We also observed an inhibition of PGE_2 production by MG pretreatment at 6 and 24 h (Figure 5A,B). This result is in agreement with the findings of Kim and co-workers (2006),[52] who showed that mast cells failed in PGD_2 generation due to the inhibition of COX-2-enzyme activity by MG. PGE_2 produced by synoviocytes and macrophages in synovial space induces increased vascular permeability and is one of the main mediators responsible for edema formation, including in ZIA,[24] and it is involved centrally in fever and pain production.[53] Studies *in vitro* and in animal models have demonstrated that PGE_2 induces bone resorption by osteoclasts[54] and regulates type II collagen synthesis and degradation and that a high concentration of PGE_2 stimulates the release of cartilage-degrading MMPs.[55] The reduction in the levels of PGE_2 in the joint after pretreatment with MG might be related to the reduction in paw edema formation and tissue damage as observed in the above results.

Methyl Gallate Inhibits the Chemotactic Activity and Adhesion of Neutrophils *in Vitro*. We have previously shown that MG can reduce the *in vivo* recruitment of neutrophils, as well as the production of pro-inflammatory cytokines and chemokines. In this part of the study, we evaluated whether MG could directly inhibit neutrophil migration upon chemotactic stimulus. The effect of MG (0.01−100 μM) on neutrophil viability was initially assessed using the lactate dehydrogenase assay, which showed that none of the tested concentrations were cytotoxic (≥92% viability at 100 μM) (Supplemental Table 1). Next, we tested the *in vitro* effect of MG on neutrophil chemotaxis. As shown in Figure 6A, CXCL-1/KC induced a significant increase in the number of migrant neutrophils compared to basal migration in the control

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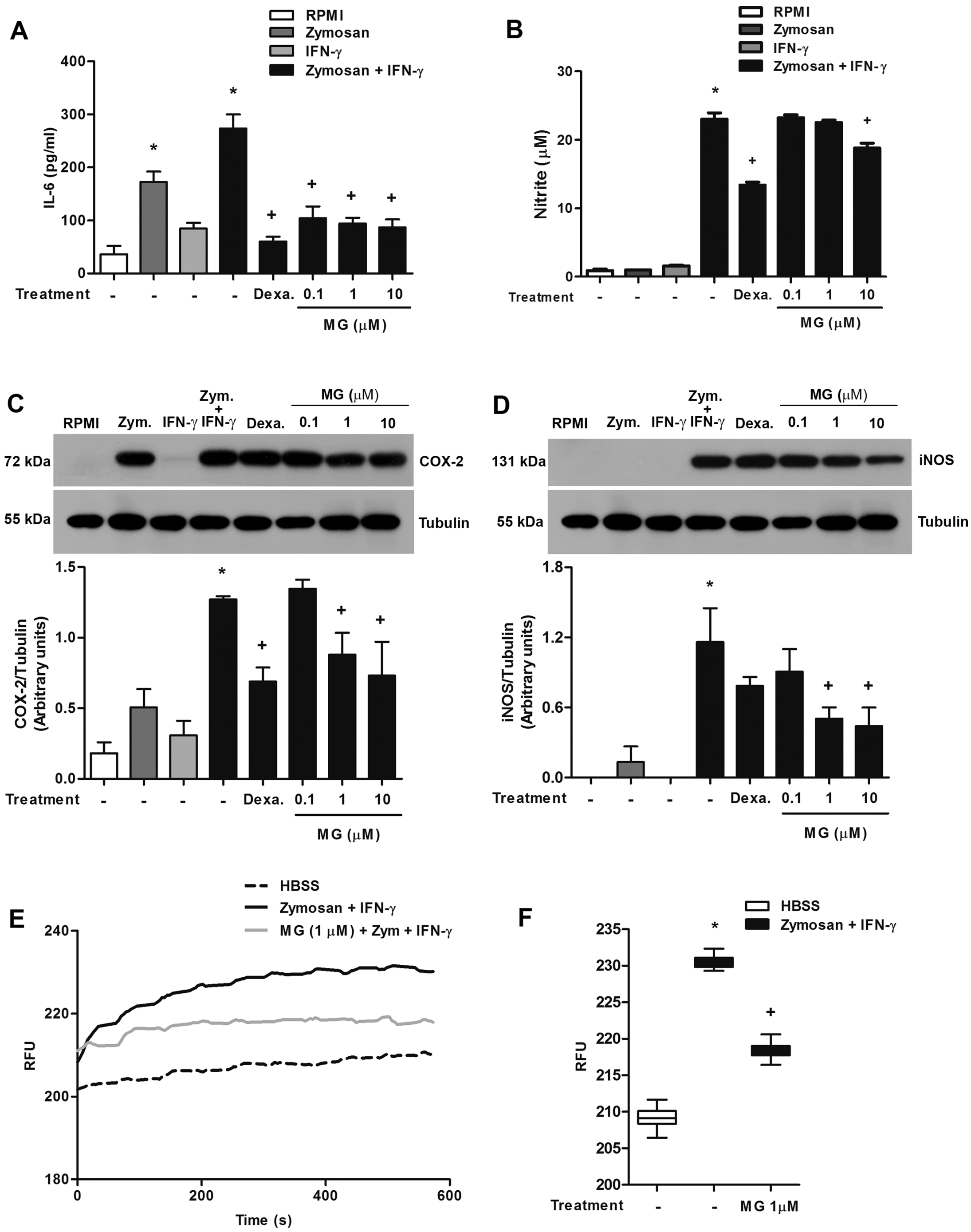

Figure 7. Methyl gallate inhibits macrophage activation. (A–D) J774A.1 macrophages were preincubated for 1 h with medium, dexamethasone (Dexa, 1 μM), or MG (0.1, 1, or 10 μM) and subjected to stimulation with zymosan (10 μg per 10^5 cells), IFN-γ (25 U/mL), or zymosan plus IFN-γ for 24 h. Levels of IL-6 (A) and NO (B) were determined in cell-free supernatant by ELISA and the colorimetric Griess test, respectively. (C and D) COX-2 and iNOS expression were quantified from cellular extracts by Western blot. (E and F) J774A.1 macrophages were preincubated for 1 h with HBSS or MG (1 μM) and stimulated with zymosan plus IFN-γ. (E) Kinetics of calcium influx of zymosan-stimulated macrophages over 600 s as measured using the FLIPR Calcium Plus assay kit and (F) means of values obtained at 360–600 s for each group after the stimulus. The data were analyzed using SOFTmax Pro. The results are presented as the means ± SEM of quadruplicate wells per group per experiment and are representative of two separate experiments [$^*p \leq 0.05$ compared to the saline group; $^+p \leq 0.05$ compared to the stimulated group (one-way ANOVA followed by Student–Newman–Keuls test)].

wells. Interestingly, MG pretreatment (0.1–10 μM) 1 h prior to CXCL-1/KC stimulation reduced neutrophil migration only at the lower concentrations (0.1 and 1 μM). Recently, Eler and co-workers (2013)[56] showed that MG has lipophilicity, which would facilitate its access to various intracellular environments. Herein, we observed that MG in the two greatest concentrations was not able to inhibit the neutrophil chemotaxis, and this unexpected effect may be due to MG binding to nonspecific sites within the cell and triggering various signaling pathways. It is evident that MG has a potent inhibitory effect on neutrophil migration; however, it seems likely that MG at higher concentrations interacts with different

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molecular targets, modifying the effects observed for lower concentrations. Dexamethasone (50 nM) also impaired neutrophil migration.

CXCL-1/KC initiates neutrophil chemotaxis through binding to G protein-coupled receptors (CXCR-1 and CXCR-2) located on the surfaces of these cells.[57] The activation of those receptors then leads to the activation of protein kinase B (PKB/Akt) and of small GTPases. This signaling regulates F-actin polymerization, which controls the direction of neutrophil migration.[58,59] A recent study showed that MG significantly reduced glioma cell migration by inhibiting the phosphorylation of Akt and the formation of focal adhesions.[15] Because MG inhibited the neutrophil chemotaxis elicited by CXCL-1, possible mechanisms involving G protein-dependent pathways should be considered in further studies.

Another important event associated with neutrophil migration is the interaction of circulating leukocytes with the vascular endothelial cells. To understand the mechanisms involved in the MG-mediated inhibition of neutrophil accumulation, we performed an *in vitro* adhesion assay. The binding of chemokines to their receptors on neutrophils almost instantaneously triggers the activation of surface integrin, which participates in the adhesion cascade of neutrophils.[60] Accordingly, neutrophils in contact with TNF-α-primed endothelial cells exhibited higher adhesion than the control group in this assay. The pretreatment of neutrophils with MG (0.1 μM, 1 h before stimulus) impaired their adhesion to endothelial cells. A similar result was obtained when neutrophils were pretreated with fucoidan, a carbohydrate that inhibits leucocyte migration by binding to L- and P-selectins, consequently inhibiting leukocyte rolling and subsequent adhesion (Figure 6B,C). This result supports the hypothesis that MG might be interfering in the signaling pathway of the chemotactic receptor, thereby avoiding neutrophil activation and the expression of adhesion molecules and consequently preventing the entry of neutrophils into sites of inflammation.

Methyl Gallate Impairs Macrophage Activation. The interaction of zymosan with resident macrophages (type A synovial lining cells) is the first step in the initiation of ZIA.[61,62] To assess the effect of MG on macrophage activation, we evaluated the production of IL-6 and NO, the expression of COX-2 and inducible nitric oxide synthase (iNOS), and the mobilization of intracellular calcium. Thus, macrophages were incubated with different concentrations of MG (0.1−10 μM) or dexamethasone (1 μM) for 1 h. The cells were stimulated with zymosan and/or IFN-γ. The cytotoxic effect of MG in macrophages was examined. As shown in Supplemental Table 2, the incubation of macrophages with MG for 24 h was not cytotoxic at any of the concentrations analyzed ($\geq$95% viability at 0.01−100 μM). Zymosan is recognized through Toll-like receptor 2 (TLR-2) and dectin-1 receptor, resulting in activation of nuclear factor-κB (NF-κB), thereby triggering the production of inflammatory cytokines and chemokines.[63,64] Classically, macrophage activation induced by the combination of a Toll ligand with IFN-γ leads to the production of cytokines such as TNF-α, IL-1β, IL-6, and IL-12 in large quantities.[65,66] In addition, activated macrophages produce reactive nitrogen species, such as the NO produced by iNOS.[67]

IL-6 is a pleiotropic cytokine that regulates the immune response, inflammation, hematopoiesis, and bone metabolism.[68] An increase in IL-6 levels was observed in both the serum and synovial fluid in patients with RA, and serum IL-6 levels were correlated with disease activity and radiographic joint damage.[69,70] Macrophages are the cells that produce inflammatory mediators, including IL-6, in the joint in large quantities.[71] Figure 7A shows that stimulation of macrophages (J774A.1) with zymosan plus IFN-γ significantly increased IL-6 production. Pretreatment with MG (0.1−10 μM) reduced the levels of IL-6 at all concentrations used. In the same way, dexamethasone pretreatment reduced IL-6 production. Unlike in neutrophils, MG at high concentrations was able to inhibit macrophage activity. One explanation for this effect is that the macrophages used are a cell line, whereas the neutrophils are primary cells. Moreover, these cells were stimulated by zymosan plus IFN-γ, which triggers different signaling pathways from those observed in the neutrophils, which were stimulated with CXCL-1/KC. Finally, we evaluated different activities in the two cell types. Thus, MG might be interacting with different molecular targets as well as modulating diverse cellular events to reduce the leukocyte infiltration in the knee joint cavity.

PGE_2 is a lipid mediator produced by COX-2, which is an inducible enzyme expressed in inflammatory conditions such as RA.[54] As previously demonstrated, MG was able to reduce PGE_2 production in ZIA. We then investigated whether this reduction resulted from the interference of MG in the expression of COX-2. Macrophages were pretreated with MG (0.1−10 μM) or with dexamethasone (1 μM) and then stimulated with zymosan and/or IFN-γ for 24 h. As shown in Figure 7C, only stimulation with zymosan plus IFN-γ significantly increased COX-2 expression. Pretreatment with MG at concentrations of 1 and 10 μM reduced the expression of COX-2, and dexamethasone inhibited the expression of this enzyme. We suggest that the reduction of *in vivo* PGE_2 production is due to decreased COX-2 expression in macrophages.

In Figure 7B and D, it shown that only stimulation with zymosan plus IFN-γ induced increases in NO production and iNOS expression. Pretreatment with MG significantly reduced NO production only at the highest concentration tested (10 μM). Moreover, MG at both 1 and 10 μM significantly inhibited iNOS expression, whereas in cells pretreated with dexamethasone there was a significant reduction only in NO production and not in enzyme expression (Figure 7B and D). The overproduction of NO induces the generation of reactive nitrogen species, which has been associated with autoimmune disease severity.[72] In the experimental model of ZIA in rats, it was observed that the production of NO occurred in association with the severe migration of neutrophils into the synovial cavity.[25] Increased iNOS expression in inflamed synovia, particularly in synovial cells and neutrophils, contributes to NO release.[73] Our results showed that MG inhibits *in vitro* NO production by macrophages (type A synoviocytes), which suggests that in ZIA macrophages might be one of the cell types involved in NO production during the exacerbation of the inflammatory process.

Dectin-1 is a β-glucan receptor expressed on the surfaces of monocytes and macrophages and is involved in zymosan recognition and internalization. Dectin-1 signaling stimulates intracellular calcium mobilization after the activation of phospholipase Cγ (PLC-γ) and the release of inositol triphosphate in the cytoplasm.[74] In macrophages, the combined stimulus of zymosan and IFN-γ was accompanied by a prolonged increase in the cytoplasmic calcium concentration, which was inhibited by pretreatment with MG (1 μM) (Figure 7E). For statistical analysis of this assay, the means of values

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obtained between 360 and 600 s for each group were plotted (Figure 7F). Calcium is a versatile intracellular messenger in eukaryotic cells, regulating many cellular processes including the cell cycle, the transport system, mobility, gene expression, and cell metabolism.[75] The presence of calcium is essential for the action of the enzyme phospholipase A_2 (PLA_2).[76] Cytosolic free calcium binds to the catalytic site of PLA_2, inducing the release of arachidonic acid from membrane phospholipids for the biosynthesis of eicosanoids.[77] The MG-induced reduction in intracellular calcium mobilization may reduce the activity of PLA_2, thereby decreasing the release of arachidonic acid. This process might partially explain the decrease in the observed PGE_2 and LTB_4 production in the synovial fluid of mice treated with MG. Indeed, we found that MG pretreatment reduced both migration and adhesion of neutrophils, as well as the production of PLA_2-dependent lipid mediators from macrophages. It is likely that MG might interfere in the signaling pathways of chemotactic receptors, consequently impairing neutrophil migration. Moreover, the inhibition of calcium influx by MG demonstrates a direct effect of MG on the modulation of macrophage functions.

The present study demonstrates for the first time the anti-inflammatory effect of MG in animal models. According to the results shown in this work, we can suggest that MG acts on signaling cascades and that its actions are not exclusively dependent on its antioxidant properties. It is possible that MG interacts with multiple molecular targets, as has been observed for numerous natural products,[78] since the activation of macrophages by zymosan triggers different signaling pathways than those observed downstream of G protein-coupled receptors. On the other hand, there is evidence that the PI3K signaling pathway is activated in response to numerous TLR stimuli and subsequently induces Akt phosphorylation.[79] Moreover, it has been demonstrated that $G\beta\gamma$ signaling contributes to macrophage membrane dynamics and zymosan phagocytosis.[80] Thus, MG may be acting on the same target via different signaling pathways, or it may have pleiotropic cellular functions. Indeed, future experiments are necessary to clarify the mechanism of action of MG.

Taken together, we demonstrated in this study that MG, a phenolic acid derivative found in various plant species, exerts an important anti-inflammatory effect on zymosan-induced inflammation, particularly in experimental arthritis. Our results showed that MG significantly reduces neutrophil migration, the production of inflammatory mediators, and the activation of cells crucial to the development of this disease. In addition, these data highlight the importance of further study on the mechanism of action of MG, reinforcing the importance of natural products as sources of therapeutics for the modulation of the inflammatory process and the control of arthritis.

■ EXPERIMENTAL SECTION

Chemicals and Reagents. Methyl gallate (98% purity) was purchased from Fluka (Sigma-Aldrich). Zymosan A, phosphate-buffered saline (PBS) tablets, perborate buffer, Tween-20, Hank's balanced salt solution (HBSS), probenecid, fucoidan, *o*-phenylenediamine dihydrochloride, Percoll, Bradford reagent, bovine serum albumin, ethylenediaminetetraacetic acid sodium salt (EDTA), phenylmethylsulfonyl fluoride (PMSF), protease inhibitors cocktail, and RPMI 1640 were all obtained from Sigma Chemical Co. (St. Louis, MO, USA). May-Grünwald and Giemsa stains were purchased from Merck (Darmstadt, Hessen, Germany). Fetal bovine serum (FBS) and gentamicin were obtained from Gibco (Grand Island, NY, USA). Polyvinylidene difluoride membrane was purchased from GE Healthcare (Chicago, IL, USA). Purified anti-murine TNF-α, CXCL1/KC, IL-1β, and IL-6 mAbs, biotinylated anti-TNF-α, CXCL-1/KC, IL-1β, and IL-6 mAbs, and recombinant TNF-α, CXCL1/KC, IL-1β, and IL-6 were obtained from R&D Systems (Minneapolis, MN, USA). LTB_4 and PGE_2 immunoassay kits were obtained from Cayman Co. (Ann Arbor, MI, USA). Goat polyclonal purified anti-COX-2 and biotin-conjugated anti-goat IgG were purchased from Santa Cruz Biotechnology (Dallas, TX, USA). Dexamethasone (Decadron; Aché, SP, Brazil) and sodium pentobarbital 3% (Hypnol; Syntec, SP, Brazil) were commercially obtained.

Animals. Male C57BL/6 mice (18−22 g) from the Oswaldo Cruz Foundation Breeding Unit (Fiocruz, Rio de Janeiro, Brazil) were kept caged with free access to food and fresh water in a room with the temperature ranging from 22 to 24 °C and a 12 h light/dark cycle. The animals were housed at the Farmanguinhos experimental animal facility unit until use. All animal care and experimental procedures performed were approved by the institution's Committee of Ethics in Animal Care and Use (CEUA) (registered under the number CEUA LW-43/14) and were performed according to the ethical guidelines of the International Association for the Study of Pain.[81]

Treatments. Animals were fasted overnight and then received MG in doses ranging from 0.7 to 70 mg/kg orally (p.o.) diluted in filtered water in a final volume of 200 μL, 1 h before zymosan stimulation. Dexamethasone (10 mg/kg, 100 μL) was administered intraperitoneally (i.p.) 1 h prior to stimulation and used as a reference inhibitor. The same volume of vehicle (200 μL) was administered orally to the control groups.

Paw Edema. To evaluate the effect of MG on paw edema induced by zymosan, C57BL/6 mice were pretreated orally with various doses of MG (1−50 mg/kg) and subjected to zymosan (i.pl., 100 μg/paw) diluted in sterile saline to a final volume of 50 μL.[82] The contralateral paw received an i.pl. injection of equal volume of sterile saline and was used as a control. Four hours after stimulus, paw edema was evaluated by plethysmography (plethysmometer 7140, Ugo Basile, Comerio, Italy). Paw edema values are expressed in microliters (μL), and the difference (Δ) between the right and left paws was taken as the edema volume.

Induction of Pleurisy. One hour after the respective treatments, pleurisy was induced in C57BL/6 mice by an i.t. injection of zymosan (100 μg/cavity) diluted in sterile saline to a final volume of 100 μL, according to the technique of Spector[83] as modified for mice by Henriques.[84] The control group received an i.t. injection of an equal volume of vehicle. The mice were euthanized 4 h after stimulus using a lethal dose of pentobarbital sodium 3% (Hypnol), and their thoracic cavities were washed with 1 mL of PBS containing EDTA (10 mM). Total leukocyte counts were performed in an automatic particle counter (Coulter Z2, Beckman Coulter Inc., Brea, CA, USA). Differential counts were performed under a light microscope (100×) using Cytospin smears (Cytospin 3, Shandon Inc., Pittsburgh, PA, USA) stained using the May-Grumwald-Giemsa method. The counts are reported as the number of cells ($\times 10^6$) per cavity. Pleural washes were centrifuged at 400*g* for 10 min to obtain the supernatant for total protein quantification by the Bradford method according to the manufacturer's instructions.

Experimental Arthritis Model. Joint inflammation was induced by the i.a. injection of zymosan (500 μg per cavity in 25 μL of sterile saline) by inserting a 27.5 G needle through the suprapatellar ligament into the left knee joint cavity, as previously described.[5,12] The contralateral knee was injected with the same volume of the vehicle and used as control. Knee joint swelling was evaluated by measurement of the transverse diameters of each knee joint with a digital caliper (Digmatic Caliper, Mitsutoyo Corporation, Japan) at different time points after stimulation. Values of knee joint thickness are expressed as the difference (Δ) between the diameters measured before (basal) and after the induction of articular inflammation in millimeters (mm). After induction of joint inflammation, mice were euthanized by a lethal dose of pentobarbital sodium 3% (Hypnol) at various time points. Knee synovial cavities were washed twice with 150 μL of PBS containing EDTA (10 mM) by the insertion of a 21 G needle into mouse knee joints, and the synovial washes were recovered

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by aspiration. The volume was inserted in stages (50 μL each time, per needle insertion). Total leukocyte counts were performed in an automatic particle counter (Coulter Z2, Beckman Coulter Inc.). Differential counts of leucocytes were performed using May-Grunwald-Giemsa-stained cytospin smears (Cytospin 3, Shandon Inc.), and the values are reported as the number of cells per cavity ($\times 10^5$). After cellular counts, the synovial washes were centrifuged at 400*g* for 10 min at 4 °C, and the supernatant was stored at −80 °C for further analysis.

Histology. Whole knee joints were obtained from six different C57BL/6 mice at 6 and 24 h after i.a. administration of zymosan or saline. The knee joints were removed, dissected, and fixed in 10% formalin for 48 h. After that, the knees were kept in 10% EDTA in PBS solution for 1−2 weeks for decalcification. Subsequently, the specimens were processed for paraffin embedding. Tissue sections (5 μm) were stained with H&E, mounted on permanent glass slides, and analyzed under an optical microscope (Olympus BX41, Olympus, Japan). Images were taken at lower magnification (10×) to allow viewing of the entire area of the synovial cavity and were graded subjectively for the following parameters: synovial hyperplasia (pannus formation; from 0 = no hyperplasia, to 3 = most severe hyperplasia); synovial infiltrate (from 0 = no inflammation, to 3 = most severe inflammation), and bleeding (from 0 = no bleeding, to 3 = most severe bleeding).[85] To eliminate potential bias, the slides were scored by two independent observers who were blinded to the experimental procedure. Scores for all parameters were subsequently summed to give an arthritis index (AI; expressed as the mean ± SEM).

Cytokine and Chemokine Measurements (ELISA). The concentrations of TNF-α, IL-1β, IL-6, and CXCL-1/KC in the knee joint washes 6 and 24 h after stimulation with zymosan were evaluated by sandwich ELISA using matched antibody pairs (Quantikine, R&D Systems, Minneapolis, MN, USA) according to the manufacturer's instructions. The results are expressed as picograms of each cytokine or chemokine per milliliter (pg/mL).

Enzymatic Immunoassay for LTB_4 and PGE_2. LTB_4 and PGE_2 levels were evaluated in cell-free synovial washes recovered from zymosan-stimulated C57BL/6 mice 6 and 24 h after zymosan (500 μg per cavity) injection. LTB_4 and PGE_2 were assayed by immunosorbent assay (EIA) according to the manufacturer's protocol (Cayman Chemical). The results are expressed as picograms of LTB_4 or PGE_2 per milliliter (pg/mL).

Preparation of Murine Bone Marrow PMN Cells. C57BL/6 mice were euthanized, and the femurs from both hind legs were dissected and removed, free of soft tissue attachments. The extreme distal tip of the bone was cut out, and each femur was washed with 2 mL of Ca^{2+}-/Mg^{2+}-free HBSS−EDTA solution. The cell suspension was then centrifuged at 400*g* for 15 min at 20 °C and resuspended in 2 mL of Ca^{2+}-/Mg^{2+}-free HBSS−EDTA. The cells were purified using Percoll discontinuous gradients (65% and 72% diluted in Ca^{2+}-/Mg^{2+}-free HBSS−EDTA). For that purpose, the cells were centrifuged at 1200*g* for 35 min at room temperature without braking, and neutrophils were recovered from the 65%/72% interface. Next, neutrophils were counted in a Neubauer chamber and identified by cytospin centrifugation followed by Wright-Giemsa coloration (∼85−90% final purity) according to the manufacturer's instructions.

Neutrophil Viability. Neutrophil viability in the presence and absence of MG was determined using the lactate dehydrogenase-based *in vitro* toxicology assay kit (Sigma). After isolation, neutrophils were plated into 96-well culture plates at a density of 1×10^5 cells/well. Then, the cells received fresh medium with or without Tween 20 (3%) or MG (0.01−100 μM) in a quadruplicate assay. After 2 h of incubation, the plate was centrifuged at 250*g* for 4 min to pellet the cells. Aliquots (100 μL) were transferred to a clean flat-bottom plate, and the lactate dehydrogenase assay mixture was added to each sample (200 μL per well) and incubated at room temperature for 20−30 min. The reaction was terminated by the addition of 1 N HCL to each well. The absorbance was read using a SpectraMax 190/Molecular Devices at a wavelength of 490 nm. The background absorbance was read at 690 nm, and this value was subtracted from the primary wavelength measurement (490 nm).

Chemotaxis Assay. Neutrophil chemotaxis was assayed in a 48-well Boyden chamber (Neuroprobe Inc., Cabin John, MD, USA). The bottom wells of the chamber were filled with 29 μL of a chemoattractant stimulus, CXCL-1/KC (250 nM) or RPMI 1640 (control), while the upper wells were filled with neutrophils (10^5 cells, 50 μL) that had been preincubated with MG (0.1, 1, and 10 μM), dexamethasone (50 nM), or medium for 1 h. The lower and upper cells were separated using a 5 μm polycarbonate filter (Nuclepore, Sigma-Aldrich). The chamber was incubated in humidified air with a 5% CO_2 atmosphere at 37 °C for 60 min, and the filter was fixed and stained using Wright-Giemsa coloration according to the manufacturer's instructions. Neutrophils that had migrated through the membrane were counted under light microscopy (100× magnification) in 15 random fields. The result is expressed as the mean of migrated neutrophils per field.

Cell Adhesion Assay. The cell adhesion assay used the murine thymic endothelioma cell line tEnd.1 cultured in RPMI 1640 medium supplemented with 10% FBS containing 0.1% gentamicin. These cells were incubated at 37 °C in a humidified incubator containing 5% CO_2 and plated onto four-well culture chambers (10^4 cells/well) (Lab-Tek chambers, Nunc, Thermo Fisher Scientific, Inc., Waltham, MA, USA) for 24 h. Before each experiment, the tEnd.1 cells were stimulated with recombinant mouse TNF-α (10 ng/mL) for 4 h. Neutrophils recovered from naïve C57BL/6 mice were pretreated with fucoidan (25 μg/mL) or MG (0.1 μM) for 1 h and then allowed to adhere to the tEnd.1 cultures (50 neutrophils/tEnd.1) for 1 h at 37 °C with shaking. Nonadherent cells were gently washed away with PBS, and the remaining cells were subsequently stained with Giemsa and visualized by light microscopy (100× magnification). The number of adhered neutrophils per tEnd.1 cell was determined by direct counting. The data are expressed as an index of adhesion (IA), calculated as follows: IA = (tEnd.1 cells with bound neutrophils)/(total number of tEnd.1 cells) × (neutrophils bound to tEnd.1 cells)/(total number of tEnd.1 cells) × 100, as previously described.[86]

Macrophage Viability. Cellular viability in the presence and absence of MG was determined using an Alamar Blue assay (Invitrogen, Carlsbad, CA, USA). Cells of the mouse cell line J774A.1 were plated into a black flat-bottomed 96-well plate at a density of 2.5×10^5 cells/well. After 24 h of incubation in a controlled atmosphere (5% CO_2, 37 °C), the cells received fresh medium with or without Tween 20 (3%) or MG (0.01−100 μM) in a quadruplicate assay. After 20 h of further incubation, 20 μL of Alamar Blue solution was added to each well, and after 4 h, the fluorescence was read using a SpectraMax M5/M5e microplate reader (Molecular Devices; λ_{exc} = 555 nm, λ_{em} = 585 nm).

Macrophage Activation and IL-6 and Nitric Oxide Production. Mouse macrophages J774A.1 (1×10^5 cells/well in RPMI with 10% FBS and gentamicin) were plated into 96-well culture plates and allowed to adhere for 24 h in a controlled atmosphere. The macrophages were incubated with dexamethasone (1 μM) or various concentrations of MG (0.1−10 μM) for 1 h. Then, the cells were stimulated with zymosan (10 μg per 10^5 cells) and/or IFN-γ (25 U/mL) for 24 h. After 24 h, the amount of IL-6 was evaluated by sandwich ELISA as described above. The amount of nitrite, a metabolite of nitric oxide, was assessed in the supernatant from the macrophages using the Griess method.[87] The absorbance was measured at 540 nm in a plate reader (Softmax Pro 190-Molecular Devices).

Western Blot. J774A.1 cells were plated into six-well plates and allowed to adhere for 24 h. The cells were then treated with MG (0.1−10 μM) or dexamethasone for 1 h before stimulation with zymosan and/or IFN-γ as described above. Whole cell lysates were obtained with lysis buffer (20 mM Tris, 10 mM NaCl, 3 mM $MgCl_2$, 0.5% NP-40, 1 mM PMSF, and protease inhibitor cocktail), and the total protein concentration of each sample was determined according to the Bradford method. Cell lysates were denatured in Laemmli buffer (50 mM Tris-HCL, pH 6.8, 1% SDS, 5% β-mercaptoethanol, 10% glycerol, and 0.001% bromophenol blue) at 90 °C for 5 min. Protein samples (30 μg of protein) were resolved by 7.5% SDS-PAGE and transferred to polyvinylidene difluoride membranes. The membranes were probed

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with antibodies against COX-2, iNOS, and α-tubulin (Santa Cruz Biotechnology). Antibodies were diluted in Tris-buffered saline containing 0.1% Tween 20 and 5% skimmed milk and incubated overnight at 4 °C. On the following day, the membranes were incubated with anti-goat, anti-rat, or anti-mouse peroxidase-conjugated secondary IgG antibody (Santa Cruz Biotechnology) for 1 h at room temperature. The blots were developed with the use of a chemiluminescent substrate (ECL Western blotting analysis system; Amersham Biosciences Buckinghamshire, UK), and images were acquired for densitometric analysis using ImageJ software.

Intracellular Calcium Measurement. The intracellular calcium was assayed using the FLIPR calcium assay kit (Molecular Devices, Sunnyvale, CA, USA), according to the manufacturer's instructions. Mouse cell line J774A.1 was plated into a black flat-bottomed 96-well plate at a density of 5×10^4 cells/well. After 24 h of incubation in a controlled atmosphere, the macrophages were incubated with probenecid (inhibitor for the anion-exchange protein) at 2.5 mM in a final volume of 100 μL. Then, MG (1 μM) was added for 1 h at 37 °C in loading buffer at a final volume of 100 μL, according to the manufacturer's protocol. Following incubation, the macrophages were either unstimulated or stimulated with zymosan (10 μg per 10^5 cells) and IFN-γ (25 U/mL). Intracellular Ca^{2+} spikes in macrophages were monitored for 10 min. The data were analyzed using SOFTmax Pro (Softmax Pro190-Molecular Devices).

Statistical Analysis. The data are reported as the mean ± SEM and analyzed statistically using analysis of variance (ANOVA) followed by the Newman-Keuls-Student test or Student's *t*-test. Values of $p \leq 0.05$ were regarded as significant.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jnatprod.5b01115.

Additional figures (PDF)

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Notes

The authors declare no competing financial interest.

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